Simulation Parameters

Actual vehicle speed (km/h)

50

Radar angle θ (degrees)

30

Signal-to-noise ratio (dB)

10

Noise random seed

0

Warning thresholds

Green/Yellow threshold (km/h)

50

Yellow/Orange threshold (km/h)

Doppler Radar Vehicle Speed Estimation Using DSP

CW Doppler radar simulation: signal generation → noise → FIR filtering → FFT peak detection → radial-speed estimation → angle correction.

Estimated Results

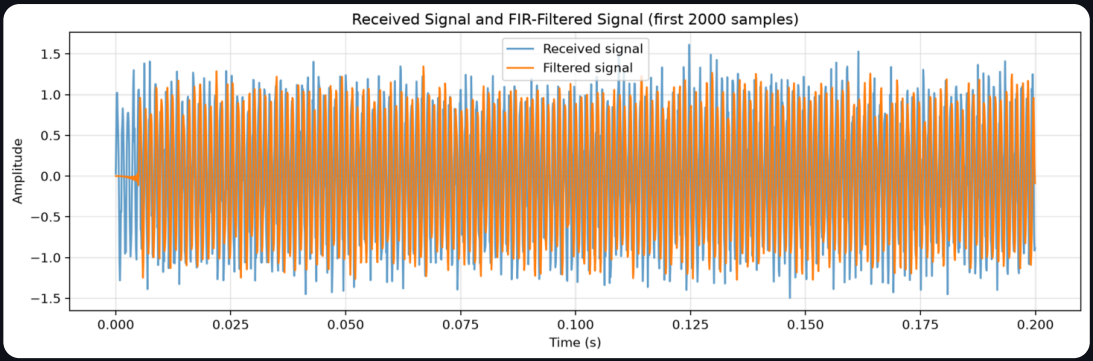
Actual speed	Radial speed (actual)	Estimated speed (c…	Doppler frequency…
50.00 …	43.30 …	50.01 …	844.0 …

Speed Warning: YELLOW — CAUTION

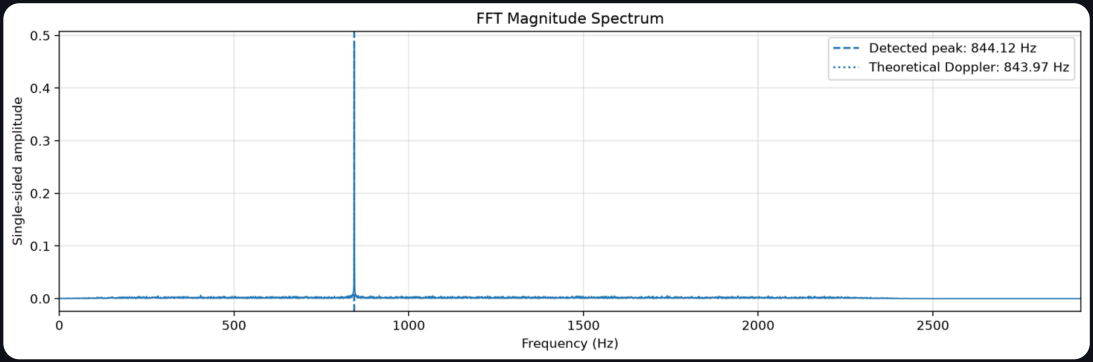
Estimated Doppler frequency: 844.12 Hz | **Absolute error:** 0.01 km/h | **Percentage error:** 0.02%

Angle correction uses the selected angle: $v = v_r / \cos(\theta)$. A single-channel CW Doppler signal does not independently determine the angle. This is a synthetic educational simulation, not measured radar data.

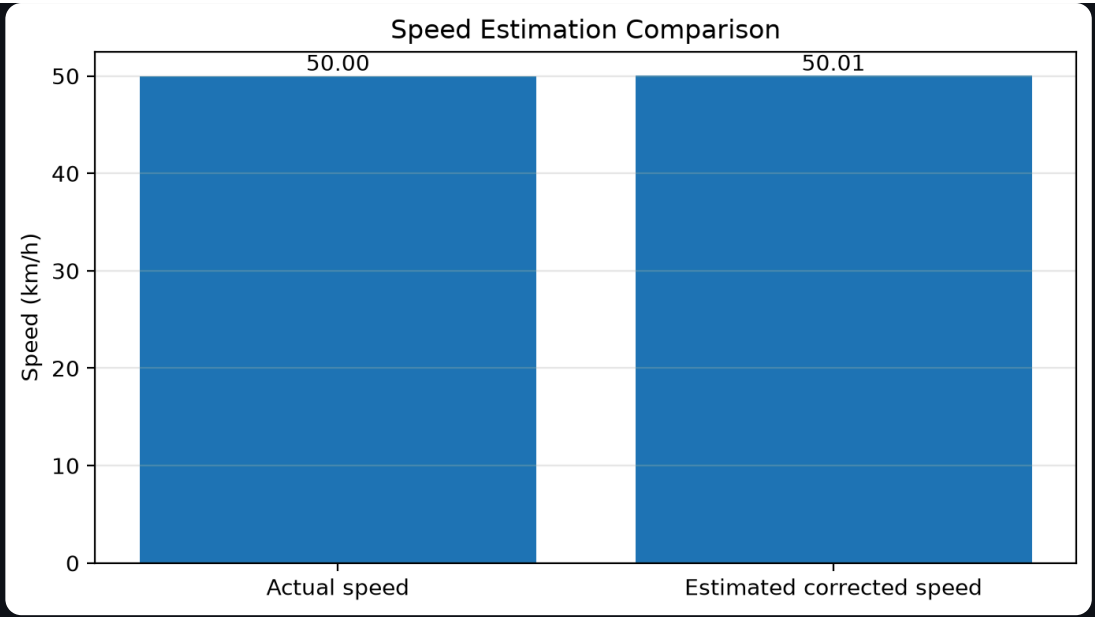
1. Time-Domain Signals



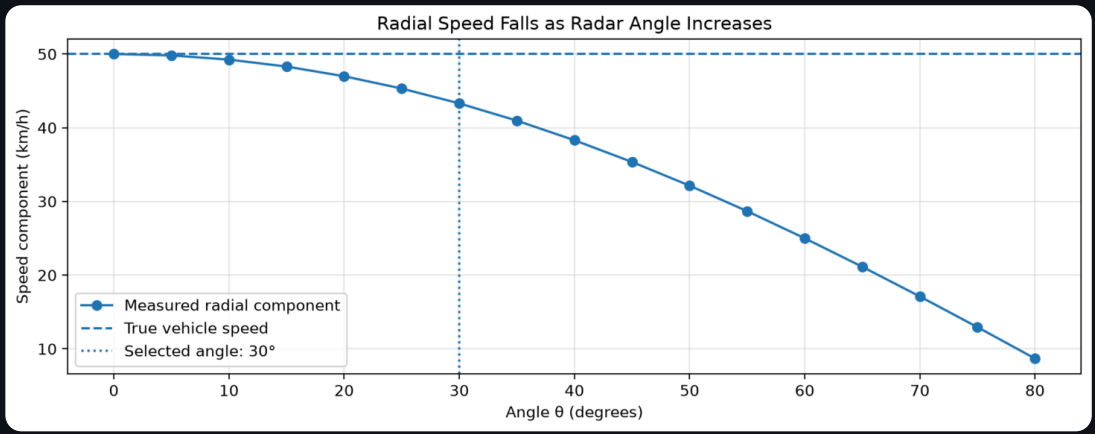
2. FFT Spectrum and Detected Doppler Peak



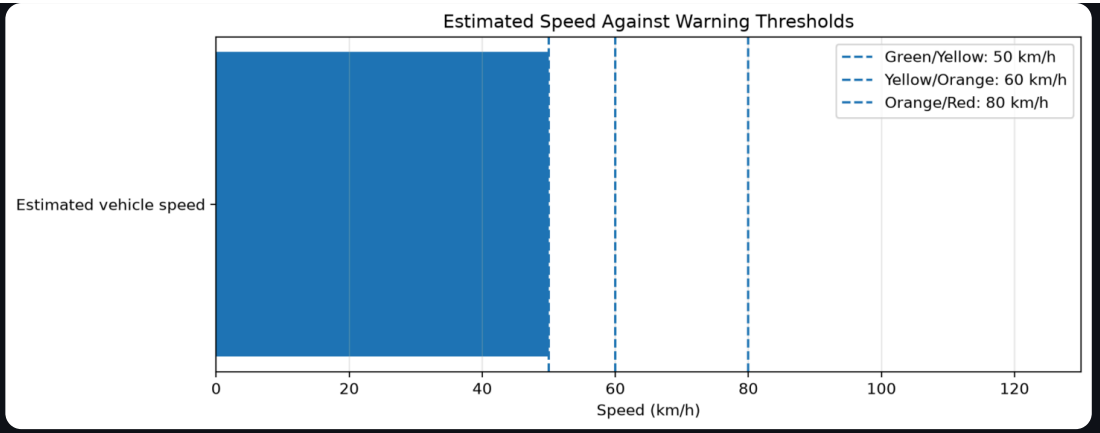
3. Actual vs Estimated Speed



4. Effect of Radar Angle



5. Speed Warning Levels



Show simulation parameters and DSP details


```
{
  "Radar carrier frequency (Hz)" : 10525000000
  "Speed of light (m/s)" : 300000000
  "Wavelength (m)" : 0.028503562945368172
  "Sampling frequency (Hz)" : 10000
  "Simulation duration (s)" : 2
  "Number of samples" : 20000
  "FFT length" : 16384
  "SNR (dB)" : 10
  "FIR filter order" : 100
  "Minimum Doppler frequency (Hz)" : 97.4537037037037
  "Maximum Doppler frequency (Hz)" : 2338.888888888889
  "Theoretical Doppler frequency (Hz)" :
  843.9738310028905
  "Detected Doppler frequency (Hz)" : 844.1162109375
```

```
  "Angle correction used" : "v = v_r / cos(30°)"
}
```

Note: The FIR filter uses a Hamming-window design. FFT-bin spacing and noise can cause small differences between actual and estimated speed. Warning thresholds are demonstration values.